MIL-G-413C
29 August 1986
SUPERSEDING
MIL-G-413B
9 October 1963
(See 6 6)

MILITARY SPECIFICATION

GLUE, MARINE

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope This specification covers material for paying wooden seams of ship decks

2 APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specification and standards The following specification and standards form a part of this specification to the extent specified herein Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation

SPECIFICATION

FEDERAL
PPP-P-704 - Pails, Metal (Shipping, Steel, 1 Through 12 Gallons).

STANDARDS

FEDERAL
FED-STD-313 - Material Safety Data Sheets Preparation and Submission of

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 8040
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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2 2 Other publications The following documents form a part of this specification to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 5 - Standard Test Method for Penetration of Bituminous Materials (DoD adopted)

D 140 - Standard Methods of Sampling Bituminous Materials (DoD adopted)

D 1084 - Standard Test Methods for Viscosity of Adhesives (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents These documents also may be available in or through libraries or other informational services)

2 3 Order of precedence In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS

3 1 Material Marine glue shall contain no materials of coal tar origin and shall contain no asbestos

3 2 Characteristics The glue shall have the characteristics shown in table I

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TABLE I Characteristics

	Minimum	Maximum	Test paragraph
Viscosity at 350 degrees Fahrenheit (°F), poises	----	8	4.4 1
Boiling point, °F	350	----	4.4 1
Penetration, centimeters			4 4 2
at 32°F	0 10	----	
at 68°F	0 30	0 50	
at 115°F	----	1 15	
Filler content, high gravity siliceous percent by weight	----	25	4 4 3
Working properties	shall pass test		4 4 4
Flexibility	shall pass test		4 4.5
Flow	shall pass test		4.4 6

3 3 Workmanship The glue shall be homogenous and free from foreign matter

3 4 Toxic products and formulations The material shall have no adverse effect on the health of personnel when used for its intended purpose Questions concerning this shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity

3 4 1 Material safety data sheet The contracting activity shall be provided with a material safety data sheet (MSDS) at the time of bid or quotation submittal The MSDS is DD Form OSHA 20 and is found in FED-STD-313 The MSDS shall be included with each shipment of the material covered by this specification (see 6 5)

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

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4 1 1 Responsibility for compliance All items must meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material

4 2 Sampling

4 2 1 Lot For purposes of sampling, a lot shall consist of marine glue from one manufactured batch offered for delivery at one time A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform

4 2 2 Sampling for examination of filled containers A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I with an acceptable quality level (AQL) of 2.5 percent defective These containers shall be examined as specified in 4 3

4 2 3 Sampling for tests Samples of glue shall be selected in accordance with ASTM D 140 These samples shall be tested as specified in 4 4

4 3 Examination of filled containers Each sample filled container shall be examined for the following defects

(a) construction of the container defective
(b) container closure defective
(c) contents of container less than specified
(d) evidence of leakage
(e) unsatisfactory workmanship

4 3 1 Rejection criteria for examination Any container having one or more defects shall be rejected If the number of defective containers in any sample exceeds the specified AQL, the entire lot represented by that sample shall be rejected

4 4 Tests

4 4 1 Viscosity and boiling point A portion of the glue shall be heated, and its viscosity shall be measured in poises at 350°F using method B of ASTM D 1084 The boiling point shall be observed during the heating process

4 4 2 Penetration The penetration of the glue shall be determined in accordance with ASTM D 5 under the following conditions

At 32°F with a load of 200 grams applied for 60 seconds
At 68°F with a load of 200 grams applied for 60 seconds
At 115°F with a load of 50 grams applied for 5 seconds

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4 4 3 Filler content An accurately weighed sample shall be gently heated in a weighed crucible until the volatile matter is expelled The glue shall then be heated at a greater rate until it is reduced to ash and the weight has become constant The percentage of high gravity siliceous filler shall then be calculated

4 4 4 Working properties

4 4 4 1 Apparatus A section of decking approximately 4 feet square shall be constructed of deck planking not less than nominal 2 inches thick by 4 to 6 inches wide The decking shall have V-shaped seams not less than 1 inch deep and not less than 3/16 inch wide at the top It shall be restrained by a framework to avoid spreading of the decking The wood shall be conditioned to a moisture content of 15 ± 3 percent It shall be exposed to a temperature of 73.5 ± 2°F for at least 24 hours The seams shall be caulked in the standard manner with oakum to a point sufficient to allow not less than 1/2 inch depth of glue

4 4 4 2 Procedure The clean, dry seams shall be payed with the glue at a temperature of 325 to 350°F, the ladle shall be held about 1 inch above the seams

4 4 4 3 Acceptance The glue shall be free running without lumps or sediment, and the payed seam shall set to a solid mass with a uniform jet-black color and glossy finish A mass containing water vapor pockets shall be unacceptable

4 4 5 Flexibility The flexibility of the glue shall be determined as follows Brush a thin coat of the hot liquid sample on a black iron panel that is not more than 0 015 inch thick Allow the coating to cool and place the panel in a suitable bath, maintained at a temperature of 0 ± 5°F, for not less than 30 minutes Remove the panel from the bath and bend the panel quickly, coated side up, through an angle of 180 degrees over a mandrel 3/8 inch in diameter The coating shall show no evidence of cracking or peeling

4 4 6 Flow The flow properties of the glue shall be determined as follows Flow a layer of the hot liquid sample onto half of one side of a black iron panel Let the sample cool in a horizontal position Scribe a line directly ahead of the sample flow and place the panel in a vertical position in an oven at 170 ± 2°F for 24 hours The glue shall not flow past or cover the scribed line

4 4 7 Rejection criteria for tests If any sample selected in accordance with 4.2.3 fails to conform to the requirements of this specification, the entire lot represented by that sample shall be rejected

4.5 Inspection of packaging Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein The inspection of commercial packaging shall be as specified in the contract or order (see 6 2)

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4 6 Toxicological formulations The contractor shall have the toxicological formulation information and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition)

5 1 Preservation Not applicable

5 2 Packing Packing shall be level A or commercial, as specified (see 6 2)

5 2 1 Level A Marine glue shall be packed in metal pails of appropriate size for the unit pack quantity specified (see 6 2) Pails shall conform to PPP-P-704, type II, class 3 or 5

5 2 2 Commercial Marine glue in the unit pack quantity specified (see 6 2) shall be packed in accordance with ASTM D 3951

5 3 Palletization When specified (see 6 2), pails of marine glue shall be palletized in accordance with MIL-STD-147

5 4 Marking In addition to any special marking required (see 6 2), level A packs shall be marked in accordance with MIL-STD-129, commercial packs in accordance with ASTM D 3951 Bar code marking shall be included, and shall be in accordance with MIL-STD-129

6 NOTES

6 1 Intended use Marine glue is intended for use in paying wooden deck seams on ships It should be employed in accordance with the procedures described in the Naval Ships Technical Manual, chapter 634

6 2 Ordering data Acquisition documents should specify the following

(a) Title, number, and date of this specification
(b) Size of containers required
(c) Inspection requirements for commercial packaging (see 4 5)
(d) Packing level required (see 5 2)
(e) Unit quantity required (see 5 2 1 and 5 2 2)
(f) Whether palletization is required (see 5 3)
(g) Special marking required (see 5 4)

6 3 Safety hazards Cautions noted in the material safety data sheet (see 3 4 1) should be observed Marine glue is heated very hot to melt it and enable it to be poured into seams Care should be taken to avoid skin burns to personnel and proper heating equipment and care should be used to avoid setting the material afire

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6.4 Subject term (key word) listing

Flexibility
Free running
Marine glue
Paying
Siliceous

6.5 Material safety data sheets Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313 The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313

6.6 Changes from previous issue Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Custodians
Navy - SH
Air Force - 99

Review activities
Army - MR, EA
Navy - SA, AS

User activity
Navy - YD

Preparing activity
Navy - SH
(Project 8040-0461)

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-G-413C	GLUE, MARINE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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